Renewable Power Hosting Capacity Assessment Considering L-index for Static Voltage Stability

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***Abstract*—The construction of new type of power system with an increasing integration of renewable energy generation has become the mainstream trend in current power system development. The rising penetration of renewable energy poses challenges to the stability of the power system, making it increasingly important to assess renewable energy hosting capacity while considering system stability constraints. This paper establishes a method for assessing renewable energy hosting capacity considering static voltage stability. Firstly, it reviews and analyzes commonly used static voltage stability indices in power systems, selecting the L-index for detailed modeling. Building upon this, a renewable energy hosting capacity assessment model incorporating the L-index is developed, and a solution strategy based on the primal-dual interior point method is proposed. Subsequently, the study analyzes the impact of considering static voltage stability on the assessment results of renewable energy hosting capacity using an enhanced IEEE 30-bus test case.**

***Keywords*—renewable energy, hosting capacity assessment, static voltage stability, L-index.**

I. INTRODUCTION

As early as the 1940s, Marković raised concerns about voltage stability in power systems. With the continuous expansion of power grid scale and increasing penetration of renewable energy, the industry has placed greater emphasis on voltage stability issues. In recent years, large-scale system incidents caused by renewable energy in modern power systems have also become more frequent.

The dynamic equations of power systems can be succinctly described as a complex set of nonlinear equations. Specifically, they can be represented by a set of nonlinear differential-difference-algebraic equations (DDAEs), which form the foundation for addressing all voltage stability issues in power systems[1].

Static voltage instability occurs when the voltage level of a power system gradually decreases as the load increases. This instability reaches a critical state when the load grows to a point where the system can no longer support it. Static voltage stability involves identifying a time-domain section of the power system's dynamic processes. Mathematically, in DDAEs, this can be simplified by fixing the difference equations in the DDAE and approximating the differentiation of state variables to zero. [5]This transformation allows the DDAE system to be described purely by algebraic equations.

Following this logic, static voltage stability is solely related to the algebraic equations describing the system. Analytical methods for assessing static voltage stability can generally be categorized into two approaches: static voltage stability indices [6][7]and methods such as the continuation power flow or direct methods [8][9]based on continuous flow analysis. Common assessment indices for static voltage stability margin include sensitivity index[10][11], eigenvalue and singular value indices[12][13], singular value indices[14], margin indices [15], V/V0 index [16], voltage stability proximity index, etc. The static voltage stability L index (L index), proposed by Kessel in 1986, belongs to the category of voltage stability proximity indices[17]. The L index is based on power flow solutions and is defined through the derivation of system network equations. It is a stability index that intuitively reflects the distance between the current operating state of the system and static voltage collapse. Due to its early proposal, the L index has been widely applied in the analysis of static voltage stability in power systems and exhibits strong accuracy. Given the excellent performance of the L index in assessing static voltage stability, this section will employ the L index to evaluate the static voltage stability of the system. The subsequent section will provide a theoretical basis by detailing the establishment and derivation process of the L index, supporting the assessment of renewable energy integration considering static voltage stability.

II. THE DEFINITION AND SIMPLIFICATION OF L-INDEX

A. The definition of the L-index

L-index requires dividing the nodes of the system into load nodes S_l and generator nodes S_g. The voltage equations of the system nodes are rearranged according to these groups:

$$\begin{bmatrix} \mathbf{I}_l \\ \mathbf{I}_g \end{bmatrix} = \begin{bmatrix} \mathbf{Y}_{ll} & \mathbf{Y}_{lg} \\ \mathbf{Y}_{gl} & \mathbf{Y}_{gg} \end{bmatrix} \begin{bmatrix} \mathbf{V}_l \\ \mathbf{V}_g \end{bmatrix} \tag{1}$$

In the equations, $\mathbf{I}_l$ and $\mathbf{I}_g$ are the current vectors of all load nodes and all generator nodes, respectively; $\mathbf{V}_l$ and $\mathbf{V}_g$ are the voltage vectors of all load nodes and all generator nodes, respectively; $\mathbf{Y}_{ll}$, $\mathbf{Y}_{lg}$, $\mathbf{Y}_{gl}$ and $\mathbf{Y}_{gg}$ are the admittance matrices between load nodes, from load nodes to generator nodes, from generator nodes to load nodes, and between generator nodes, respectively.

1) Model transformation

Substituting $Z_{ll} = Y_{ll}^{-1}$ into equation (1) yields the following expression:

$$\begin{bmatrix} \mathbf{V}_l \\ \mathbf{I}_l \end{bmatrix} = \begin{bmatrix} \mathbf{Z}_{ll} & -\mathbf{Z}_{ll}\mathbf{Y}_{lg} \\ \mathbf{Y}_{gl}\mathbf{Z}_{ll} & \mathbf{Y}_{gg} - \mathbf{Y}_{gl}\mathbf{Z}_{ll}\mathbf{Y}_{lg} \end{bmatrix} \begin{bmatrix} \mathbf{I}_l \\ \mathbf{V}_g \end{bmatrix} \tag{2}$$

$$\dot{V}_j = \sum_{i,j \in S_l} Z_{ji} \dot{I}_i - \sum_{\substack{j \in S_l \\ k \in S_g}} Z_{jk} Y_{jk} \dot{V}_k \tag{3}$$

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In the equations, $\dot{V}_j \in \mathbf{V}_{\mathrm{l}}\left(j \in S_l\right)$ and $\dot{V}_k \in \mathbf{V}_g\left(k \in S_g\right)$ represent the voltage vectors of load node *i* and generator node *k*, respectively; $\dot{I}_i \in I_{\mathrm{l}}\left(i \in S_l\right)$ denotes the current vector of load node *i*; $Z_{ji} \in \mathbf{Z}_{\mathrm{ll}}\left(i, j \in S_l\right)$ denotes the element at the *i*-th row and *j*-th column of the impedance matrix of load nodes; $Z_{jk} \in Z_{\mathrm{lg}}\left(j \in S_l, k \in S_g\right)$ represents the impedance between load node *j* and generator node *k*; $Y_{jk} \in Y_{\mathrm{lg}}\left(j \in S_l, k \in S_g\right)$ denotes the admittance between load node *i* and generator node *k*.

2) Substitution of variables

Introduction of the load participation factor matrix $\mathbf{F}_{\mathrm{lg}} = -\mathbf{Z}_{\mathrm{ll}}\mathbf{Y}_{\mathrm{lg}}$,thus defining the equivalent corrected voltage $\dot{V}_{oj}^{+}$:

$$\dot{V}_{oj}^{+} = -\sum_{k \in S_g} F_{jk}\dot{V}_k \tag{4}$$

In the equation, F_{jk} represents the element at the *j*-th row and *k*-th column of matrix $\mathbf{F}_{\mathrm{lg}}$.

Introducing the transformation admittance matrix Y_{jj}^{+}, transformed power S_j^{+}, and equivalent power correction S_j^{corr}, where:

$$Y_{jj}^{+} = \frac{1}{Z_{jj}} \tag{5}$$

$$S_j^{+} = S_j + S_j^{corr} = S_j + \left(\sum_{\substack{i,j \in S_l \\ i \neq j}} \frac{Z_{ij}^{*} S_i}{Z_{jj}^{*}\dot{V}_i}\right)\dot{V}_j \tag{6}$$

In the equation, S_j represents the complex power at load node *j*. From equations (4) to (6), it can be seen that the voltage at load node *j* is mainly influenced by the power S_j at node *j* itself and the equivalent power contributions S^{corr} from other load nodes to node *j*.

Substituting $\dot{V}_{oj}^{+}, \mathbf{F}_{\mathrm{lg}}, Y_{jj}^{+}, S_j^{+}$ and S_j^{corr} into equation (3), we obtain:

$$1 + \frac{\dot{V}_{oj}^{+}}{\dot{V}_j^{*}} = \frac{S_j^{+*}}{Y_{jj}^{+*}V_j^2} \tag{7}$$

3) Define the L index

According to equation (7), define the static voltage stability L index as:

$$\mathrm{L}_j = \left|1 + \frac{\dot{V}_{oj}^{+}}{\dot{V}_j^{*}}\right| = \left|\frac{S_j^{+*}}{Y_{jj}^{+*}V_j^2}\right| \tag{8}$$

The static voltage stability L index, after simplification, can be written as:

$$\mathrm{L}_j = \frac{\left|\sum_{i,j \in S_l} \frac{Z_{ji}^{*} S_i}{\dot{V}_i}\right|}{V_j} \tag{9}$$

B. The simplification of the L index

The variation of loads in a power system has a minimal impact on the phase angle difference at nodes. Therefore, the phase angle part of the voltage in equation (9) can be neglected, resulting in the simplified static stability L index:

$$\mathrm{L}_j^{'} = \frac{1}{V_j}\sqrt{\left(\sum_{i \in \alpha_L} \frac{P_i R_{ji} + Q_i X_{ij}}{V_i}\right)^2 + \left(\sum_{i \in \alpha_L} \frac{Q_i R_{ji} - P_i X_{ij}}{V_i}\right)^2} \tag{10}$$

In practical transmission networks, the reactance of the power system is much larger than the resistance. Based on the characteristics of real power grids, equation (10) can be further simplified as follows [18]:

$$\mathrm{L}_j^{\cdot} = \frac{1}{V_j}\sqrt{\left(\sum_{i \in \alpha_L} \frac{Q_i X_{ij}}{V_i}\right)^2 + \left(\sum_{i \in \alpha_L} \frac{-P_i X_{ij}}{V_i}\right)^2} \tag{11}$$

$\mathrm{L}_j \in (0,1)$, when L index values are maintained between 0 and 1, L index serves as a measure of the system's static voltage stability. When L index approaches 0, the system is at the boundary of static voltage instability, indicating higher static voltage stability.

III. THE ASSESSMENT MODEL FOR RENEWABLE ENERGY HOSTING CAPACITY CONSIDERING THE L INDEX.

A. Objective function

Maximizing renewable energy output as the objective function:

$$\max\left(\sum_{i=1}^{N_p} P_i^p + \sum_{i=1}^{N_w} P_i^w\right) \tag{12}$$

In the equation, N_p and N_w denote the total numbers of photovoltaic units and wind turbine units in the system, respectively. P_i^p represents the active power output of the *i*-th photovoltaic unit, and P_i^w represents the active power output of the *i*-th wind turbine unit.

B. Constraints

1) Node power balance constraint

$$\begin{cases} P_i^p + P_i^w + P_i^g - P_i^l = \sum_{j=1}^{N_n}\left(G_{ij}V_iV_j\cos\theta_{ij} + B_{ij}V_iV_j\sin\theta_{ij}\right) \\ Q_i^g - Q_i^l = -\sum_{j=1}^{N_n}\left(B_{ij}V_iV_j\cos\theta_{ij} - G_{ij}V_iV_j\sin\theta_{ij}\right) \end{cases} \tag{13}$$

In the equation, N_n represents the total number of nodes in the system. P_i^g and Q_i^g denote the active and reactive power generated by conventional generator units at node *i*, respectively. P_i^l and Q_i^l represent the active and reactive power demand at node *i*, respectively. G_{ij} and B_{ij} denote the real and imaginary parts of the admittance matrix element at row *i* and column *j*, respectively. V_i represents the voltage magnitude at node *i*, and θ_{ij} represents the phase angle difference of voltage between nodes *i* and *j*.

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2) Constraint on active power output of conventional generator units

$$\begin{cases} P^{g}_{\min} \le P^{g}_{i} \le P^{g}_{\max} \\ Q^{g}_{\min} \le Q^{g}_{i} \le Q^{g}_{\max} \end{cases} \tag{14}$$

In the equation, $P^{p}_{\max}$ and $P^{w}_{\max}$ represent the theoretical maximum power generation of photovoltaic and wind power respectively.

3) Constraints on renewable energy output

$$\begin{cases} 0 \le P^{p}_{i} \le P^{p}_{\max} \\ 0 \le P^{w}_{i} \le P^{w}_{\max} \end{cases} \tag{15}$$

In the equation, $P^{p}_{\max}$ and $P^{w}_{\max}$ represent the theoretical maximum power generation of photovoltaic (PV) and wind power respectively.

4) Transmission power constraints

$$P^{\min}_{ij} \le P_{ij} \le P^{\max}_{ij} \tag{16}$$

In the equation, P_{ij} represents the active power flow from node *i* to node *j*. $P^{\min}_{ij}$ and $P^{\max}_{ij}$ denote the minimum and maximum allowable active power flow on branch , respectively.

5) Node voltage magnitude constraints

$$V^{\min}_{i} \le V_{i} \le V^{\max}_{i} \tag{17}$$

In the equation, $U^{\min}_{i}$ and $U^{\max}_{i}$ denote the minimum and maximum allowable values of the voltage at node *i*, respectively.

6) L-index constraints

$$\begin{cases} 0 \le \mathrm{L}_{j} \le \mathrm{L}_{\mathrm{MAX}} \\ \mathrm{L}_{j} = \dfrac{1}{V_{j}} \sqrt{\left(\sum_{i \in \alpha_{L}} \dfrac{Q_{i} X_{ij}}{V_{i}}\right)^{2} + \left(\sum_{i \in \alpha_{L}} \dfrac{-P_{i} X_{ij}}{V_{i}}\right)^{2}} \end{cases} \tag{18}$$

In the equation, L_{j} represents the static voltage stability index of load node *j* in the system.

IV. CASE STUDY

A. Case Study Introduction and Convergence Verification

The improved IEEE 30-node network structure used in this section is depicted in the following figure. In this scenario setup, conventional generator units are connected to nodes 1, 2, 13, 22, 23, and 27. Additionally, a photovoltaic power plant is connected to node 5, and a wind energy plant is connected to node 28. The base voltage is set at 115 kV, and the base capacity is 100 MVA.

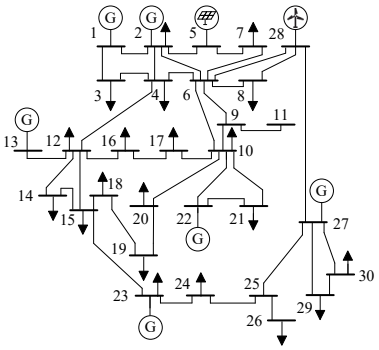


Fig. 1. Improved IEEE30 bus system

The relevant boundary conditions for the mathematical optimization model are set as follows:

TABLE I. BOUNDARY CONDITION SETTINGS

Parameters	Values (p.u.)
Reactive Power Limits for Generator at Node 1	1.5000/-0.2000
Reactive Power Limits for Generator at Node 2	0.6000/-0.2000
Reactive Power Limits for Generator at Node 13	0.4470/-0.1500
Reactive Power Limits for Generator at Node 22	0.6250/-0.1500
Reactive Power Limits for Generator at Node 23	0.4000/-0.1000
Reactive Power Limits for Generator at Node 27	0.4870/-0.1500
Active Power Limitation for Photovoltaic Units	2.0000/0.0000
Active Power of Wind Turbines	2.2000/0.0000
Branch Transmission Power Limitation	0.5
Voltage Amplitude Limitation	1.05/0.95
L index limitation	0.4

The computational basis of this research plan is the Core(TM) i7-10700F CPU, 2.9GHz desktop computer, used for solving the model. MATLAB programming is employed to optimize and solve the renewable energy integration capacity assessment model considering static voltage stability (hereinafter referred to as the L-index integration model) established in using the original dual interior-point method.

Using the original dual interior-point method, different L_{max} values were selected to iteratively solve the aforementioned mathematical optimization model. Ultimately, with five different L_{max} values selected, the algorithm converged successfully in all cases, with the maximum number of iterations being 29 and the minimum 20.

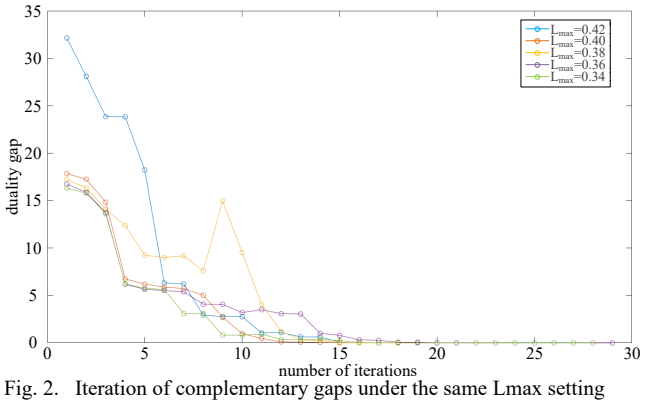


Fig. 2. Iteration of complementary gaps under the same Lmax setting

B. Comparison

1) Consideration of the L-index Absorption Model and Traditional Absorption Model

Based on the mathematical model established earlier, set up control groups for the traditional absorption model and the L-index absorption model. In Scenario 1, the L-index absorption model considers the same mathematical setup as earlier, with a static voltage stability L-index constraint where Lmax=0.4. In Scenario 2, the traditional absorption model does not consider the static voltage stability L-index constraint, omitting equations (7) and (8) from the previously established model. The absorption results of the two scenarios are analyzed as follows:

a) Static Voltage Stability

After evaluating the L-index in the absorption models, it is observed that the overall L-index in the L-index absorption model is generally lower compared to the traditional absorption model. Additionally, the peak L-index in the L-index absorption model evaluation is 0.3982 p.u., significantly lower than the peak L-index of 0.4417 p.u. in the traditional absorption model. This demonstrates that the absorption model considering the L-index can achieve better static voltage stability. The calculated average voltage in the L-index absorption model is 0.9983 p.u., whereas in the traditional absorption model, it is 0.9935 p.u., further supporting the superior static voltage stability of the L-index absorption model.

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The distribution of L-indexes after evaluation in both the L-index absorption model and the traditional absorption model is shown in the following figure:

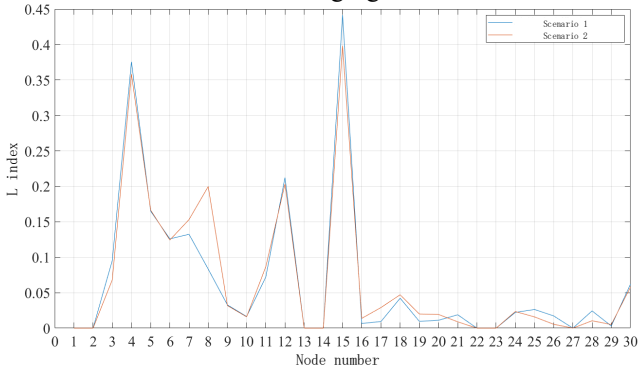


Fig. 3. Distribution of L indicators

The voltage comparison between the absorption model considering the L-index and the traditional absorption model is as follows:

TABLE II. VOLTAGE AMPLITUDE COMPARISON

Node	Scenario1 Voltage (p.u.)	Scenario2 Voltage (p.u.)	Node	Scenario1 Voltage (p.u.)	Scenario2 Voltage (p.u.)
1	1.0000	1.0000	16	0.9876	0.9861
2	1.0000	1.0000	17	0.9880	0.9885
3	0.9982	0.9962	18	0.9773	0.9767
4	0.9979	0.9955	19	0.9752	0.9752
5	1.0426	1.0500	20	0.9795	0.9798
6	0.9986	0.9974	21	0.9961	0.9963
7	1.0064	1.0085	22	1.0000	1.0000
8	0.9893	0.9883	23	1.0000	1.0000
9	0.9967	0.9970	24	0.9879	0.9876
10	0.9957	0.9968	25	0.9906	0.9906
11	0.9967	0.9970	26	0.9728	0.9728
12	0.9940	0.9902	27	1.0000	1.0000
13	1.0000	1.0000	28	1.0201	1.0198
14	0.9851	0.9815	29	0.9805	0.9805
15	0.9866	0.9847	30	0.9691	0.9691

b) Hosting capacity of renewable power

The total hosting capacity of renewable power in the system is 3.5583 p.u., and the penetration rate is 84.72%; The traditional assessment model has a hosting capacity of 3.8919 p.u. and a penetration rate of 92.66%. From the data comparison in Table 2.5.3, it can be seen that when considering the L indicator, due to the limitations of the L indicator, the system's renewable power penetration level will be significantly reduced.

The comparison of the hosting capacity situation evaluated by the L-index assessment model and the traditional assessment model is shown in the table below:

TABLE III. COMPARISON OF HOSTING CAPACITY SITUATION

Node	Scenario1 (p.u.)		Scenario2(p.u.)	
	Hosting capacity	Rate	Hosting capacity	Rate
PV	1.8657	93.28%	1.6127	80.64%
Wind	2.0262	92.10%	1.9456	88.44%
Total	3.8919	92.66%	3.5583	84.72%

2) The hosting capacity under different Lmax settings.

Based on the assessment model considering the L index set in the previous section, we solve the assessment model under different L_{max} settings using the aforementioned solution strategy. The comparison of hosting capacity situations under different L_{max} settings is presented in the table below:

TABLE IV. HOSTING CAPACITY UNDER DIFFERENT LMAX SETTINGS

Scenario	L_{max}	PV(p.u.)		Wind(p.u.)	
		Hosting capacity	Rate	Hosting capacity	Rate
1	0.40	1.6127	80.64%	1.9456	88.44%
2	0.38	1.5550	77.75%	1.8948	86.13%
3	0.36	1.4997	74.99%	1.8365	83.48%
4	0.34	1.4380	71.90%	1.7802	80.92%
5	0.32	1.4024	70.12%	1.7280	78.55%
6	0.30	1.3624	68.12%	1.6480	74.91%
7	0.28	1.3315	66.58%	1.5789	71.77%
8	0.26	1.2881	64.41%	1.5009	68.22%
9	0.24	1.2488	62.44%	1.4330	65.14%
10	0.22	1.1949	59.75%	1.3672	62.15%
11	0.20	1.1595	57.98%	1.2967	58.94%

Based on the table above, plot the scatter diagram as follows:

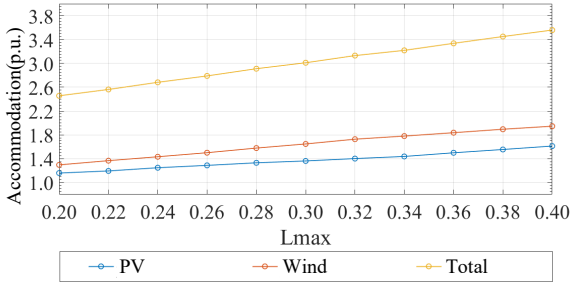


Fig. 4. Scatter plot of absorption capacity under different Lmax settings

From data analysis, it can be seen that when L_{max} decreases from 0.4 to 0.2, the static voltage stability of the system improves accordingly. However, the total hosting capacity of renewable power decreases from 3.5583 p.u. to 2.4562 p.u. For every 0.02 decrease in L_{max}, the average total hosting capacity decreases by 0.1102 p.u. The total hosting capacity of the system is positively correlated with the size of L_{max}. For every 0.02 decrease in L_{max}, the average photovoltaic hosting capacity decreases by 0.0453 p.u., and the average wind power hosting capacity decreases by 0.0649 p.u.

3) The Relationship between Renewable Energy Site Selection, Static Voltage Stability, and Renewable Energy hosting capacity

On the basis of the hosting capacity assessment model considering the L index set in the previous text, different renewable energy access scenarios are set up to analyze the impact of different renewable energy access scenarios on the L index and hosting capacity. L_{max}=0.40 is set, the assembled capacity of photovoltaic generators is 2.0000p. u., and the hosting capacity of wind turbines is 2.2000p. u.

According to the analysis results of the traditional assessment model in the graph, it can be seen that the L index distribution of nodes 15 and 4 in the traditional assessment model is 0.4417 and 0.3756, which are the two nodes with the highest L index among all load nodes; The L indicators of nodes 14 and 29 are 0.0000 and 0.0030, which are the two nodes with the lowest L indicators among all load nodes. Based on this, establish the following scenarios:

TABLE V. HOSTING CAPACITY UNDER DIFFERENT LOCATIONS

Scenario	PV node	Wind node	Hosting capacity		
			PV	Wind	Total
1	4	15	2.0000p.u.	1.4658p.u.	3.4658p.u.
2	15	4	1.4229p.u.	2.2000p.u.	3.6229p.u.
3	14	29	0.8220p.u.	0.4467p.u.	1.2687p.u.
4	29	14	0.4467p.u.	0.8235p.u.	1.2702p.u.
5	15	14	0.9015p.u.	0.6202p.u.	1.5217p.u.
6	14	15	0.4561p.u.	1.1235p.u.	1.5796p.u.

The assessment model considering the L index was used for calculation and analysis, and the situation of the six scenarios mentioned above is shown in the above figure. Scenario 1 and Scenario 2 are connected to renewable energy by the two nodes with the highest L index (i.e. the weakest static voltage stability of the system) under traditional model analysis; Scenario 3 and Scenario 4 are connected to renewable energy by the two nodes with the lowest L index

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(i.e. the strongest static voltage stability) under traditional model analysis; Scenario 5 and Scenario 6 add renewable energy to the two nodes with the highest and lowest system L indicators.

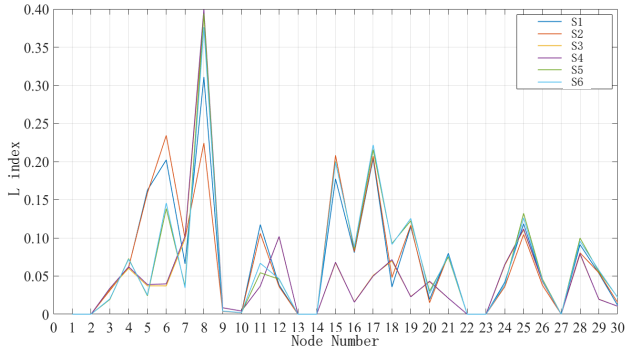


Fig. 5. Distribution of L-index nodes in different scenarios

After considering the assessment model of the L index for calculation and analysis, it is found that when renewable energy is added to the node with the weakest static voltage stability, the system's hosting capacity is larger. When renewable energy is added to the node with the strongest static voltage stability, the absorption capacity of renewable energy will be significantly reduced.

V. SUMMARY

This article first studies the static voltage stability indicators and related analysis and calculation methods of the power system. The L indicator is selected to evaluate the static voltage stability of the studied system. Subsequently, based on the static voltage stability L index, a renewable energy hosting capacity assessment model considering the L index was established. By comparing the evaluation results of the system's absorption capacity under different L_{max} settings, it was found that the evaluation results of the system's absorption capacity were basically positively correlated with the value of L_{max}, that is, the larger the L_{max} value, the better the evaluation results of the system's absorption capacity. The conclusion can be achieved that when renewable energy units are connected to nodes with higher L index, the system's absorption rate and amount are higher. This conclusion can have certain reference value for renewable energy site selection.

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